

1 Vector bundles

Let X be a topological space and $\pi : E \rightarrow X$ a (complex) vector bundle over X . By this we mean

1. a locally trivial bundle in the sense that for each $x \in X$ there exists an open neighborhood $U_x \subset X$ and a homeomorphism $h_U : \pi^{-1}(U) \rightarrow U \times \mathbb{C}^n$ such that $p_U \circ h_U(e) = \pi(e)$, where $p_U : U \times \mathbb{C}^n \rightarrow U$ is the projection in the first factor. We call h_U a trivialization of E over U .
2. For each $x \in X$ the fiber $\pi^{-1}(x)$ is a vector space and for any trivialization $h_U : \pi^{-1}(U) \rightarrow U \times \mathbb{C}^n$, the map $p'_U \circ h_U|_{\pi^{-1}(x)} \rightarrow \mathbb{C}^n$, where $p'_U : U \times \mathbb{C}^n \rightarrow \mathbb{C}^n$ (the projection on the second factor), is an isomorphism.

We say that the vector bundle is C^∞ (holomorphic) if all the manifolds and maps are C^∞ (holomorphic). The dimension of the fibers is called the rank of the vector bundle and in the case the dimension is one the vector bundle is said to be a line bundle. Morphisms between vector bundles E and F are maps $\varphi : E \rightarrow F$ which map linearly fibers into fibers. An isomorphism is a morphism which is a diffeomorphism whose restriction to each fiber is an isomorphism between vector spaces. A trivial vector bundle over X is a vector bundle isomorphic to $X \times \mathbb{C}^n$. The definition of real vector bundles is the same with $\mathbb{R}$ substituted for $\mathbb{C}$.

1.1 Transition Cocycles

Given a vector bundle $\pi : E \rightarrow X$ over X and trivialisations $h_{U_i} : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^n$ (in the case of real vector bundles) defined over a covering $X = \bigcup U_i$ one can define the maps

$$h_{U_i} \circ h_{U_j}^{-1} : (U_i \cap U_j) \times \mathbb{C}^n \rightarrow (U_i \cap U_j) \times \mathbb{C}^n$$

which have the form

$$h_{U_i} \circ h_{U_j}^{-1}(x, v) = (x, g_{ij}(x)v)$$

where $g_{ij} : U_i \cap U_j \rightarrow GL(n, \mathbb{C})$, $GL(n, \mathbb{R})$ in the case of real vector bundles, (are called transition functions of the vector bundle associated to the covering $\bigcup U_i$. If the vector bundle is C^∞ (holomorphic) then the transition functions are C^∞ (holomorphic).

The transition functions satisfy a cocycle condition, namely, on $U_i \cap U_j \cap U_k$ we have

$$g_{ij}g_{jk} = g_{ik}.$$

Conversely, given a family of transition functions satisfying the cocycle condition one can construct a vector bundle. To see that, we construct the disjoint union $\tilde{E} = \bigcup_i U_i \times \mathbb{C}^n$ with projection $\tilde{\pi}(x, v)_i = x$. To obtain the vector bundle we quotient this space by the equivalence relation $(x, v)_i \equiv (y, w)_j$ if and only if $x = y$ and $v = g_{ij}w$.

Let $\varphi : E \rightarrow E$ be an isomorphism whose projection on the base is the identity. Let $\bigcup U_i$ be a covering and chose vector bundle trivialisations h_U and h'_U on the source and on the target, respectively. We can write then

$$h'_{U_i} \circ \varphi \circ h_{U_i}^{-1}(x, v) = (x, \varphi_i(x)v)$$

where $\varphi_i(x) \in GL(n, \mathbb{C})$. On $U_i \cap U_j$ we obtain on one hand

$$h'_{U_j} \circ h_{U_i}^{-1}(x, v) = (x, g'_{ji}v)$$

and, on the other hand,

$$h'_{U_j} \circ h'^{-1}_{U_i}(x, v) = h'_{U_j} \circ \varphi \circ \varphi^{-1} \circ h'^{-1}_{U_i}(x, v) = h'_{U_j} \circ \varphi \circ h^{-1}_{U_j} \circ h_{U_j} \circ h^{-1}_{U_i} \circ h_{U_i} \circ \varphi^{-1} \circ h'^{-1}_{U_i}(x, v)$$

That is

$$g'_{ji} = \varphi_j g_{ji} \varphi_i^{-1}.$$

In particular, if φ is the identity we obtain the description of all possible transition functions over a fixed cover.

Example 1.1 *The tangent space of a surface:*

Let (U_i, φ_i) be an atlas of a surface S . We define transition cocycles as the Jacobian matrices $g_{ij} : U_i \cap U_j \rightarrow GL(2, \mathbb{R})$ defined by

$$g_{ij}(x) = D(\varphi_i \circ \varphi_j^{-1})(x).$$

The corresponding vector bundle associated to these transition functions is called the vector bundle of S and is denoted by TS .

Example 1.2 *The holomorphic tangent space of a surface:*

If the surface has a complex structure we can use a holomorphic atlas (U_i, z_i) to define the rank one complex vector space, called holomorphic tangent bundle, with transition functions

$$g_{ij} = \frac{\partial z_j}{\partial z_i}.$$

Example 1.3 *The cotangent space of a surface:*

Let (U_i, φ_i) be an atlas of a surface S . We define transition cocycles by taking the transpose of the Jacobian matrix

$$g_{ij}^*(x) = D(\varphi_i \circ \varphi_j^{-1})^*(x).$$

The corresponding vector bundle is denoted by T^*S .

Example 1.4 *The canonical line bundle over a surface or holomorphic cotangent space:*

From the holomorphic atlas (U_i, z_i) we define the rank one complex vector space, called holomorphic cotangent bundle, with transition functions

$$g_{ij} = \frac{\partial z_i}{\partial z_j}.$$

Example 1.5 *Line bundles over $\mathbb{C}P^1$*

The line bundles over $\mathbb{CP}^1$ can be described by the transition cocycles defined over the covering $U_0 = \mathbb{C}$ and $U_1 = \mathbb{C}^* \cup \{\infty\}$. We let $g_{01} = z^n$ and denote by $\mathcal{O}(n)$ the corresponding holomorphic line bundle. Observe that from

$$dz_1 = \frac{-1}{z_0^2} dz_0,$$

the canonical line bundle over $\mathbb{CP}^1$ is identified to $\mathcal{O}(-2)$.

Remark : All vector bundles over $\mathbb{CP}^1$ can be obtained using those building blocks. In fact, a theorem of Birkhoff and Gothendieck shows that any holomorphic vector bundle of rank n over $\mathbb{CP}^1$ is isomorphic to $\mathcal{O}(n_1) \oplus \cdots \oplus \mathcal{O}(n_k)$, for some $n_i \in \mathbb{Z}$.

Suppose S is a fixed Riemann surface and consider the space of line bundles over S with isomorphism between bundles defined as isomorphism of vector bundles whose induced map on the base is the identity. Given two line bundles L_1 and L_2 one can form their product $L \otimes L'$ by defining the transition functions $g_{ij}g'_{ij}$. The inverse of a line bundle L with transition functions g_{ij} is denoted by L^* it has transition functions g_{ij}^{-1} . This product is clearly commutative and defines a group structure on the space of line bundles modulo isomorphisms.

Definition 1.6 *The Picard group $\text{Pic}(S)$ associated to a Riemann surface is the abelian group of all holomorphic line bundles modulo isomorphisms.*

1.2 Sections of vector bundles

Definition 1.7 *A section of a vector bundle $\pi : E \rightarrow X$ is a map $s : X \rightarrow E$ such that $\pi \circ s(x) = x$. If the vector bundle is C^∞ (holomorphic) then we can consider C^∞ (holomorphic) sections.*

In local trivialisations over a covering $\bigcup U_i$, a section is given by functions $f_i : U_i \rightarrow \mathbb{C}^n$ satisfying the compatibility condition

$$f_i(x) = g_{ij}f_j(x)$$

on $U_i \cap U_j$ where g_{ij} are the transition functions.

The space of sections of a vector bundle is a vector space.

Definition 1.8 *The space of holomorphic sections of a holomorphic vector bundle E over a complex manifold M is denoted by $H^0(M, E)$.*

1.2.1 Meromorphic sections

Let S be a Riemann surface and $\pi : E \rightarrow S$ a holomorphic vector bundle.

Definition 1.9 *A meromorphic section of E is a holomorphic section $s : X \setminus D \rightarrow E$ where $D \subset S$ is a discrete set such that for each $p \in D$ there exists a chart $z : U \rightarrow \mathbb{C}$ with $U \cap D = \{p\}$ satisfying*

- $z(p) = 0$,

- there exists $k \geq 0$ such that $z^k(x)s(x)$ is the restriction of a holomorphic section over U to $U \setminus \{p\}$.

We set $-\text{ord}_p(s)$ to be the minimum k as above and call it the order of the meromorphic section at p .

In local coordinates this means that the n -tuple of functions f_i defined over each U_i are meromorphic. Meromorphic functions are simply meromorphic sections of the trivial holomorphic bundle $S \times \mathbb{C}$.

If a meromorphic section $s : S \rightarrow E$ vanishes at $p \in S$, consider a chart $z : U \rightarrow \mathbb{C}$ vanishing at p as above. Define $\text{ord}_p(s) = k$ to be the positive integer such that $s = z^k g$ where g is an n -tuple of holomorphic functions over U with $g(p) \neq 0$. Those definitions are clearly independent on the chosen charts.

1.2.2 Operators

Let $E_1 \rightarrow M_1$ and $E_2 \rightarrow M_2$ be two vector bundles.

Definition 1.10 A linear map $P : H^0(M_1, E_1) \rightarrow H^0(M_2, E_2)$ is called an operator.

If $E \rightarrow M$ is a holomorphic vector space we define the Cauchy-Riemann map

$$\bar{\partial}_E : H^0(M, E) \rightarrow H^0(M, E)$$

by fixing a trivialization e_i over a neighborhood U and writing any section over U as $s(z) = \sum f_i(z)e_i$. Define

$$\bar{\partial}_E(s) = \bar{\partial}_E(\sum f_i(z)e_i) = \sum \bar{\partial}(f_i(z))e_i.$$

By choosing another trivialization so that $e'_j = g_{ji}e_i$ with g_{ji} holomorphic, we observe the definition does not depend on the trivialization.

Examples

1. $\mathcal{E}^p(U)$ of p -forms on an open set U in a manifold M . Those are sections of the bundle $\Lambda^p T^*(U)$. In local coordinates $(x^1, \dots, x^n)$ they are described by

$$\sum \varphi_{i_1 \dots i_p}^x dx^{i_1} \wedge \dots \wedge dx^{i_p}$$

such that in the intersection with another coordinate chart $(y^1, \dots, y^n)$ we have

$$\sum \varphi_{i_1 \dots i_p}^y dy^{i_1} \wedge \dots \wedge dy^{i_p}$$

with

$$\varphi_{i_1 \dots i_p}^y = \sum \varphi_{j_1 \dots j_p}^x \frac{\partial x^{j_1}}{\partial y^{i_1}} \dots \frac{\partial x^{j_p}}{\partial y^{i_p}}.$$

The de Rham operator is defined in local coordinates as

$$d\left(\sum \varphi_{i_1 \dots i_p}^x dx^{i_1} \wedge \dots \wedge dx^{i_p}\right) = \sum_{j=1}^n \sum \frac{\partial \varphi_{i_1 \dots i_p}^x}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}.$$

On a surface a 1-form is written

$$\varphi = \varphi_1 dx^1 + \varphi_2 dx^2,$$

and

$$d\varphi = \left(\frac{\partial \varphi_2}{\partial x^1} - \frac{\partial \varphi_1}{\partial x^2} \right) dx^1 \wedge dx^2.$$

Observe that on a surface $\mathcal{E}^p = 0$ for $p \geq 3$.

2. $\mathcal{Z}^p(U)$ of closed p -forms on an open set U in a manifold M , that is, forms $\varphi \in \mathcal{E}^p(U)$ such that $d\varphi = 0$.
3. $\mathcal{E}^{p,q}(U)$ of (p, q) -forms on an open set U in a complex manifold M . In local coordinates $(z^1, \dots, z^n)$ they are given by

$$\sum \varphi_{i_1 \dots i_p, j_1 \dots j_q}^z dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_q}$$

The Dolbeault operator is defined in local coordinates as

$$\begin{aligned} & \bar{\partial} \left(\sum \varphi_{i_1 \dots i_p, j_1 \dots j_q}^z dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_q} \right) \\ &= \sum_{k=1}^n \sum \frac{\partial \varphi_{i_1 \dots i_p, j_1 \dots j_q}^z}{\partial \bar{z}^k} d\bar{z}^k \wedge dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_q}. \end{aligned}$$

On a Riemann surface a 1-form in $\varphi \in \mathcal{E}^1(U)$ can be written using a complex coordinate $z : U \rightarrow \mathbb{C}$:

$$\varphi = \varphi_z dz + \varphi_{\bar{z}} d\bar{z},$$

A form in $\mathcal{E}^{1,0}$ is written in local coordinates as

$$\varphi = \varphi_z dz$$

and the Dolbeault operator is written in local coordinates as

$$\bar{\partial}\varphi = -\frac{\partial \varphi_z}{\partial \bar{z}} dz \wedge d\bar{z}.$$

Observe that on a surface $\mathcal{E}^{p,q} = 0$ for $p \geq 2$ or $q \geq 2$.

4. $\mathcal{Z}^{p,q}(U)$ of $\bar{\partial}$ -closed (p, q) -forms on an open set U in a complex manifold M . That is, $\varphi \in \mathcal{Z}^{p,q}(U)$ if $\bar{\partial}\varphi = 0$. On a Riemann surface, because of the previous item, $\mathcal{Z}^{p,q}(U) = \mathcal{E}^{p,q}(U)$ if $q = 1$. On the other hand, a closed form in $\mathcal{E}^{1,0}(U)$ satisfies

$$\bar{\partial}(\varphi_z dz) = \frac{\partial \varphi_z}{\partial \bar{z}} d\bar{z} \wedge dz = 0.$$

That means that φ_z is holomorphic.

5. $\Omega^p(U)$ of holomorphic p -forms on an open set U in a complex manifold M . Those are p forms which in local coordinates are written

$$\varphi = \sum \varphi_{i_1 \dots i_p}^z dz^{i_1} \wedge \dots \wedge dz^{i_p}$$

with $\varphi_{i_1 \dots i_p}^z$ holomorphic.

1.3 Divisors and line bundles

In this section we describe holomorphic line bundles by divisors.

1.3.1 Divisors on Riemann surfaces

Definition 1.11 *Let S be a Riemann surface. A divisor on S is a locally finite linear combination*

$$D = \sum s_i z_i$$

where $s_i \in \mathbb{Z}$ and $z_i \in S$.

Locally finite meaning that the set of points $\{z_i\}$ is discrete in S .

The set of divisors $\text{Div}(S)$ on a fixed Riemann surface forms an abelian group generated by its points.

The divisor is said to be effective if $s_i \geq 0$ (we write $D \geq 0$). This defines a partial order by writing $D_1 \geq D_2$ if $D_1 - D_2 \geq 0$.

To a meromorphic section s of a holomorphic vector bundle over a Riemann surface one can associate the divisor

$$\text{div}(s) = \sum \text{ord}_p(s) p$$

where the sum is over all zeros and poles of the section. In particular if s is holomorphic the divisor is effective.

Definition 1.12 *Two divisors are linearly equivalent if their difference is the divisor of a meromorphic function.*

The divisor of a meromorphic function f is called a principal divisor and is denoted by (f) .

Definition 1.13 *The degree of a divisor $D = \sum s_i z_i$ is $\deg D = \sum s_i$.*

The degree defines a homomorphism $\deg : \text{Div}(S) \rightarrow \mathbb{Z}$. In the next paragraph we show that from a divisor we obtain a line bundle. In fact the Chern class of the line bundle is determined by the degree.

1.3.2 Divisors $\rightarrow$ linebundles

Proposition 1.14 *Given a divisor $D = \sum s_i a_i$, $1 \leq i \leq n$ over a Riemann surface S one can associate a line bundle $L(D)$ and a meromorphic section s such that $\text{div}(s) = D$.*

Proof. Chose charts $z_i : U_i \rightarrow \mathbb{C}$ for $1 \leq i \leq n$ such that $p_i \in U_i$, $z_i(p_i) = 0$, and $U_i \cap U_j = \emptyset$ for $i > 0$, $i \neq j$. Let $U_0 = S \setminus \{p_1, \dots, p_n\}$. Let $f_i(z) = z_i^{s_i}(z)$, for $z \in U_i$, $1 \leq i \leq n$ and $f_0(z) = 1$ for $z \in U_0$. We define transition functions by $g_{ij} = f_i/f_j$ which are clearly holomorphic in the intersections $U_i \cap U_j$. They also satisfy the cocycle condition and therefore define a holomorphic bundle which we denote $L(D)$. Moreover, the functions f_i defined on U_i match up to form a global meromorphic section f , as $f_i = g_{ij} f_j$ on the intersections, such that $\text{div}(s) = D$. One can check that using different choice of charts one gets an isomorphic line bundle.

□

Example 1.15 Let $D = n0$ be the divisor defined on $\mathbb{CP}^1$ with support on the point $0 \in \mathbb{C} \subset \mathbb{CP}^1$. Given the covering $U_0 = \mathbb{C}$ and $U_1 = \mathbb{C}^* \cup \{\infty\}$ (we changed notations with respect to the proposition above) consider the meromorphic functions z^n on U_0 and 1 on U_1 . We obtain transition functions

$$g_{01} = \frac{z^n}{1} = z^n.$$

We conclude that $L(n0) = \mathcal{O}(n)$.

Observe that if s_1 and s_2 are two meromorphic sections of a line bundle over S , there exists a meromorphic function f defined on S such that $s_2 = fs_1$. Therefore the divisors defined by them differ by a principal divisor. That gives the motivation for the next proposition.

Proposition 1.16 Two divisors are linearly equivalent if and only if their associated line bundles are isomorphic.

Proof.

Let D and D' be two divisors. If $L(D)$ is isomorphic to $L(D')$ then there exists $\varphi : L(D) \rightarrow L(D')$ and therefore $\varphi(s_D) = fs_{D'}$ for a meromorphic function f . As $\text{div}(\varphi(s_D)) = \text{div}(s_D)$ we obtain that $\text{div}(D) - \text{div}(D') = (f)$.

Exercise 1.17 Prove the converse.

□

In order to show that the correspondence between classes of divisors and classes of line bundles is one to one we need to show that there exists a non-trivial meromorphic section of a given line bundle.

2 Differential calculus on a surface

2.1 Exterior differentiation

Recall the exterior differentiation of a 0-form f defined on a surface is, in local coordinates (x_1, x_2) , given

$$df = \frac{\partial f}{\partial \tilde{x}_1} dx_1 + \frac{\partial f}{\partial \tilde{x}_2} dx_2.$$

For a 1-form $\alpha = \varphi_1 dx_1 + \varphi_2 dx_2$ it is

$$d\alpha = \left(\frac{\partial \varphi_2}{\partial x_1} - \frac{\partial \varphi_1}{\partial x_2} \right) dx_1 \wedge dx_2.$$

2.2 Integration

Given a differential form α on a surface S and a piece-wise smooth curve $c : [0, 1] \rightarrow S$ we define the integral

$$\int_c \alpha$$

using local charts $\varphi : U \rightarrow \mathbb{C}$ with coordinates (x, y) . That is, suppose $Im(c) \subset U$ and $\alpha = \varphi_1 dx + \varphi_2 dy$ then

$$\int_c \alpha = \int (\varphi_1 \dot{x} + \varphi_2 \dot{y}) dt.$$

If $Im(c)$ is not contained in a single coordinate chart we use a partition $0 = t_0 < t_1 < \dots < t_n = 1$ of $[0, 1]$ so that each $c([t_i, t_{i+1}])$ is contained in a coordinate chart. Clearly this definition does not depend on the chart because if $(\tilde{x}, \tilde{y})$ are different coordinates then $\alpha = \varphi_1 dx + \varphi_2 dy = \left(\varphi_1 \frac{\partial x}{\partial \tilde{x}} + \varphi_2 \frac{\partial y}{\partial \tilde{x}} \right) d\tilde{x} + \left(\varphi_1 \frac{\partial x}{\partial \tilde{y}} + \varphi_2 \frac{\partial y}{\partial \tilde{y}} \right) d\tilde{y}$ and therefore by the chain rule

$$\varphi_1 \dot{x} + \varphi_2 \dot{y} = \varphi_1 dx + \varphi_2 dy = \left(\varphi_1 \frac{\partial x}{\partial \tilde{x}} + \varphi_2 \frac{\partial y}{\partial \tilde{x}} \right) \dot{\tilde{x}} + \left(\varphi_1 \frac{\partial x}{\partial \tilde{y}} + \varphi_2 \frac{\partial y}{\partial \tilde{y}} \right) \dot{\tilde{y}}$$

and the integrals are the same.

Proposition 2.1 *Let α be a closed form and c, c' be homotopic curves between two points x_0, x_1 on a surface. Then $\int_c \alpha = \int_{c'} \alpha$.*

Proof. By Stokes theorem (see next section). $\square$

Theorem 2.2 *On a simply connected surface every closed 1-form α is exact. That is, there exists a function F (called a primitive of α) such that $\alpha = dF$. Two primitives differ by a constant.*

Proof. It follows from the previous proposition by defining $F(x) = \int_{x_0}^x \alpha$ as the integral does not depend on the path of integration. $\square$

In general, if S is a Riemann surface and $\pi : \tilde{S} \rightarrow S$ is its universal cover, then $\int_{\tilde{c}} \pi^* \alpha = \int_c \alpha$. So if α is a 1-form on a Riemann surface S we can compute its integral

$$\int_c \alpha = F(\tilde{c}(1)) - F(\tilde{c}(0))$$

where $\tilde{c}$ is a lift of c to the universal cover of S and F is a primitive of $\pi^* \alpha$.

Remark: Let Γ be the group of Deck transformations of the cover $\pi : \tilde{S} \rightarrow S$. If F is a primitive of the form $\pi^* \alpha$ then $F \circ \gamma$ is also a primitive because $d(F \circ \gamma) = d\gamma^* F = \gamma^* dF = \gamma^* \pi^* \alpha = (\pi \gamma)^* \alpha = \pi^* \alpha$. As two primitives differ by a constant we obtain that $F \circ \gamma = F + a_\gamma$.

Definition 2.3 *Let α be a closed one-form defined on a surface S . The period map associated to α is the homomorphism*

$$\pi_1(S, x_0) \rightarrow \mathbb{C} \quad \text{given by} \quad c \rightarrow \int_c \alpha.$$

Let Γ be the group of Deck transformations of the cover $\pi : \tilde{S} \rightarrow S$ and F a primitive of α defined on $\tilde{S}$, then the image of the period map is given by the set $\{ a_\gamma \mid \gamma \in \Gamma \}$ where a_γ are defined in the remark above. This can be seen easily if we interpret an element of Γ as a closed curve c with lift $\tilde{c}$. Then

$$\int_c \omega = F(\tilde{c}(1)) - F(\tilde{c}(0)) = F(\gamma \tilde{c}(0)) - F(\tilde{c}(0)) = a_\gamma.$$

Theorem 2.4 *Suppose a closed differential form has all periods zero. Then it has a primitive.*

Proof. Construct explicitly the primitive as $F(z) = \int_{z_0}^z \alpha$ where z_0 is a point in S . This function is well defined as the periods are null. $\square$

Corollary 2.5 *If ω is a closed holomorphic form on a compact Riemann surface such that the associated period map is zero then $\omega = 0$.*

Proof. By the previous theorem the form ω has a primitive. It is holomorphic on a compact Riemann surface therefore constant. $\square$

2.2.1 Differential 2-forms

Using local coordinates $z = x + iy$ we may write a differential 2-form as

$$\alpha = f dx \wedge dy = \frac{i}{2} f dz \wedge d\bar{z},$$

where f is a function.

If $\varphi : V \rightarrow U$ is a diffeomorphism, recall the change of variable formula

$$\int \int_U f dx dy = \int \int_V \varphi^* f du dv$$

which can be written more explicitly as

$$\int \int_U f dx dy = \int \int_V f \circ \varphi \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv$$

where $\frac{\partial(x,y)}{\partial(u,v)}$ is the Jacobian determinant. In the case φ is a biholomorphism we have

$$\int \int_U f dz \wedge d\bar{z} = \int \int_V f \circ \varphi \left| \frac{dz}{dw} \right|^2 dw \wedge d\bar{w}$$

To define the integral of a 2-form on a Riemann surface we use a partition of unit subordinated to a cover by charts. The fundamental theorem we will use is the following version of Stokes theorem.

Theorem 2.6 (Stokes Theorem) *Let α be a smooth 1-form defined on a neighborhood of a domain Ω with piecewise smooth boundary $\partial\Omega$ contained in a surface.*

$$\int_{\partial\Omega} \alpha = \int_{\Omega} d\alpha.$$

2.2.2 Cauchy's integral formula

We will admit the following integral formula.

Theorem 2.7 *Let $\Omega \subset \mathbb{C}$ be a connected open domain whose boundary is a union of finitely many C^1 Jordan curves. Let $f \in C^1(\Omega)$. Then, for $z \in \Omega$,*

$$2\pi i f(z) = \int_{\partial\Omega} \frac{f(\zeta)}{\zeta - z} d\zeta + \int_{\Omega} \frac{\partial f(\zeta)/\partial \bar{\zeta}}{\zeta - z} d\zeta \wedge \bar{\zeta}.$$

2.2.3 The residue theorem

Let ω be a meromorphic 1-form which is not identically null. Let $p \in X$ and $z : U \rightarrow \mathbb{C}$ be a chart such that ω is holomorphic on $U \setminus \{p\}$. We define the residue of ω at p as

$$\text{res}_p(\omega) = \frac{1}{2\pi i} \int_{\gamma} \omega$$

where γ is a curve with winding number 1 around p contained in U . It is easy to see that this integral is well defined. It can be computed using a Taylor expansion; write, using local coordinates, $\omega = f(z)dz$ where $f(z)$ has a pole at p and the residue is simply the coefficient of the term $\frac{1}{z}$ in the Taylor expansion. If we change the local chart then $\omega = g(w)dw = f(z)\frac{dw}{dz}dz$ and the residue is the same.

Proposition 2.8 *If X is compact then*

$$\sum_{p \in X} \text{res}_p(\omega) = 0$$

Proof. Stokes theorem. Suppose $D = \{p_i\}_{1 \leq i \leq n}$ are the poles of ω . Choose non-intersecting neighborhoods U_i containing each p_i with boundary γ_i and compute

$$\sum_i \int_{\gamma_i} \omega = - \int_{X - \bigcup U_i} d\omega = 0$$

because $d\omega = \bar{\partial}\omega + \partial\omega = 0$. □

Proposition 2.9 *If X is compact and f is a meromorphic function, then the degree of the divisor $\text{div}(f)$ is zero.*

Proof. This follows from the proposition above and the fact that $\deg(f) = \sum_{p \in X} \text{res}_p(\omega)$ for $\omega = df/f$. □

3 Homology and Cohomology

3.0.4 The de Rham complex

The de Rham complex over a surface S is

$$0 \rightarrow \mathbb{R} \rightarrow \mathcal{E}^0(S) \xrightarrow{d} \mathcal{E}^1(S) \xrightarrow{d} \mathcal{E}^2(S) \xrightarrow{d} 0.$$

where $\mathcal{E}^0(S) = C^\infty(S)$ is the space of C^∞ functions on S , $\mathcal{E}^i(S)$ is the space of i -forms on S and d is the exterior differentiation. Poincaré's lemma says that the sequence is locally exact. The cohomology groups measure how much the sequence is far from being exact. Observe that the space of closed or exact forms are vector spaces.

Definition 3.1 *The i -th cohomology group $H^i(S, \mathbb{R})$, of the surface S is the quotient of the space of closed i -forms by the space of exact i -forms.*

Observe that $\dim H^0(S, \mathbb{R})$ is the number connected components of S . In fact the space of exact 0-forms is formed by the trivial vector space of null functions.

In order to compute $H^1(S, \mathbb{R})$ we will introduce the singular homology. A singular p -simplex is a differential map from a p -simplex to S . We will write sometimes (P) for a singular 0-simplex, (P_1, P_2) for a singular 1-simplex and (P_1, P_2, P_3) for a singular 2-simplex. Fix now an abelian group G (we will mostly use $\mathbb{Z}$ or $\mathbb{R}$). A p -chain is a finite linear combination of singular p -simplices with coefficients in G . The space of p -chains will be noted C_p (with a convention that $C_{-1} = \{0\}$). There exists a boundary operator $\partial : C_p \rightarrow C_{p-1}$ satisfying $\partial^2 c = 0$ for any chain c . It is defined on singular simplices by the formulas (using the obvious notation for the restriction of maps to the boundary of a simplex)

$$\partial(P) = 0 \quad \partial(P_1, P_2) = (P_2) - (P_1) \quad \partial(P_1, P_2, P_3) = (P_2, P_3) - (P_1, P_3) + (P_1, P_2)$$

and extended by linearity to all chains.

A chain c is called a cycle if $\partial c = 0$ and a boundary if there exists a chain $\tilde{c}$ such that $\partial \tilde{c} = c$. We define

Definition 3.2 *The p -th homology group, $H_p(S, G)$ is the quotient of the space of cycles, Z_n , by the space of boundaries, B_n .*

If the surface S is connected $\dim H_0(S, \mathbb{R}) = 1$. If S is compact, orientable and connected then $\dim H_2(S, \mathbb{R}) = 1$. To compute $H_1(S, \mathbb{Z})$, we will invoke van Kampen theorem, that describes the first homology as the abelianization of the fundamental group:

$$H_1(S, \mathbb{Z}) = \frac{\pi_1(S, z)}{\{[a, b] \mid a, b \in \pi_1(S, z)\}}.$$

Using the generators a_i, b_i , $1 \leq i \leq g$ for a compact surface of genus g we obtain that $H_1(S, \mathbb{Z}) = \mathbb{Z}^{2g}$. The generators a_i, b_i , viewed as a basis of $H_1(S, \mathbb{Z})$ are also called a canonical basis for the homology. It follows from general theorems on the homology that we also have $H_1(S, \mathbb{R}) = \mathbb{R}^{2g}$.

The relation between homology and cohomology is essentially given by Stokes theorem on a chain c :

$$\int_{\partial c} \omega = \int_c d\omega.$$

Lemma 3.3 *If ω is closed and c_1 and c_2 are two homologous chains then*

$$\int_{c_1} \omega = \int_{c_2} \omega.$$

Proof. By hypothesis $c_2 - c_1 = \partial C$. Apply Stokes theorem. □

This lemma shows that the bilinear map in the following theorem is well defined.

Theorem 3.4 *Let S be a compact orientable surface of genus p . The bilinear map $H_1 \times H^1 \rightarrow \mathbb{R}$ defined by*

$$(c, \omega) \rightarrow \int_a \omega$$

is non-degenerate.

Proof. The fact that $(\cdot, \omega)$ is non-zero follows from the fact that if all periods are null, the form ω is null. On the other hand, given an element $c \in H_1$ we construct a form such that $(c, \omega) \neq 0$ in the following two lemmas. $\square$

Suppose S is orientable. Let γ be simple closed curve in S . We consider an annulus A containing γ and let A^- be the left side and A^+ the right side. Let f be a function with compact support on A^- which is one on A^- intersected with a neighborhood of γ . Define then $\eta_\gamma = df$. Even if f is not continuous, η_γ is C^∞ 1-form. On the other hand η_γ is not exact in general. The form η_γ is dual to γ in the sense of the following lemma.

Lemma 3.5 *Let ω be a closed 1-form. Then*

$$\int_\gamma \omega = \int_S \eta_\gamma \wedge \omega.$$

Proof. We compute

$$\int_S \eta_\gamma \wedge \omega = \int_{A^-} df \wedge \omega = \int_{A^-} d(f\omega) - \int_{A^-} f d\omega = \int_\gamma f\omega = \int_\gamma \omega.$$

$\square$

Remark: Using notation of the next section we write $\int_\gamma \omega = (\omega, *\eta_\gamma)$.

Lemma 3.6 *Let a_i, b_i be an homology basis. Then*

$$\int_{a_i} \eta_{a_j} = \int_{b_i} \eta_{b_j} = 0 \quad \int_{a_i} \eta_{b_j} = - \int_{b_i} \eta_{a_j} = \delta_{ij}.$$

Proof. The first equality follows from the previous lemma. For the second one, we compute in the case that a, b are two loops intersecting once at a point with orientation given by the tangent vectors to a and b at the point of intersection in that order. we denote by f_b a function associated to the loop b with support in A_b^- as before. We obtain

$$= \int_a \eta_b = \int_a df_b = 1.$$

The last equality follows from the explicit form of the function f_b at the intersection point; it corresponds to the integration on a closed interval $[0, 1]$ of the derivative of a function such that $f(0) = 0$ and $f(1) = 1$. $\square$

3.0.5 The Dolbeault complex

Recall the Cauchy-Riemann operator $\bar{\partial} = \frac{1}{2}(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y})$ defined on functions on an open subset $U \subset \mathbb{C}$. It is better understood in the guise of an operator:

$$\bar{\partial} : \mathcal{C}^\infty(U) \rightarrow \mathcal{E}^{0,1}(U)$$

given by $f \rightarrow \frac{\partial f}{\partial \bar{z}} d\bar{z}$ so that we need the following result (local solvability of the Cauchy-Riemann equation): for each $g \in \mathcal{C}^\infty(U)$ there exists $V \subset U$ and $f \in \mathcal{C}^\infty(V)$ such that

$$\frac{\partial f}{\partial \bar{z}} = g.$$

on V . A stronger result is true:

Proposition 3.7 (Dolbeault's lemma) *Let $\Omega \subset \mathbb{C}$ be an open subset and $g \in C^\infty(\Omega)$. Then there exists a function $f \in C^\infty(\Omega)$ such that*

$$\frac{\partial f}{\partial \bar{z}} = g.$$

Proof.

There are two cases:

1. In the first case we suppose g of compact support. An explicit solution is given in terms of the integral formula

$$f(z) = \frac{1}{2\pi i} \int \int_{\mathbb{C}} \frac{g(w)}{w - z} dw \wedge d\bar{w}.$$

The integral is well defined as can be seen by using polar coordinates $w - z = re^{i\theta}$ so that $\frac{1}{w-z} dw \wedge d\bar{w} = \frac{-2ir}{re^{i\theta}} dr \wedge d\theta$. Because g is of compact support, the integration is made in a sufficiently large rectangle and therefore we may differentiate under the integral sign. We obtain making the change w for $w - z$

$$\frac{\partial f(z)}{\partial \bar{z}} = \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi i} \int \int_{|w| > \epsilon} \frac{\partial g(z+w)}{\partial \bar{z}} \frac{1}{w} dw \wedge d\bar{w}.$$

So

$$\begin{aligned} \frac{\partial f(z)}{\partial \bar{z}} &= \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi i} \int \int_{|w| > \epsilon} \frac{\partial}{\partial \bar{w}} \left(\frac{g(z+w)}{w} \right) dw \wedge d\bar{w} = - \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi i} \int \int_{|w| > \epsilon} d \left(\frac{g(z+w)}{w} dw \right) \\ &= \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi i} \int_{|w|=\epsilon} \frac{g(z+w)}{w} dw = g(z). \end{aligned}$$

2. If $\text{supp } g \subset \Omega$ is not compact we construct an exhaustion sequence of compact sets K_n ($K_n \subset \text{Int}(K_{n+1})$ with $\Omega \setminus K_n$ having no relatively compact component) and cut-off functions φ_n with $\varphi_n|_{K_n} = 1$ and $\varphi_n|_{K_{n+1}} = 0$. We solve

$$\frac{\partial f_n}{\partial \bar{z}} = \varphi_n g.$$

We would like to make sense of

$$f = f_n + (f_{n+1} - f_n) + (f_{n+2} - f_{n+1}) + \cdots$$

As this sum might not converge we modify each term by a holomorphic function using Runge's theorem: As $f_{m+1} - f_m$, $m \geq 1$, is holomorphic on a neighborhood of K_n there exists a holomorphic function h_m on Ω such that

$$|f_{m+1} - f_m - h_m| < \frac{1}{2^m}$$

on K_m . We redefine the sum to be

$$f = f_n + (f_{n+1} - f_n - h_n) + (f_{n+2} - f_{n+1} - h_{n+1}) + \cdots$$

Now the sum is uniformly convergent on K_m for each $m \geq n$ so f is well defined on Ω . Moreover we immediately see that on each K_m f solves the equation.

□

Remark 3.8 *On an n -dimensional complex manifold we have the following exact sequence*

$$0 \rightarrow \mathcal{O} \longrightarrow \mathcal{C}^\infty \xrightarrow{\bar{\partial}} \mathcal{E}^{0,1} \xrightarrow{\bar{\partial}} \mathcal{E}^{0,2} \dots,$$

and more generally

$$0 \rightarrow \Omega^{p,q} \longrightarrow \mathcal{E}^{p,q} \xrightarrow{\bar{\partial}} \mathcal{E}^{p,q+1} \xrightarrow{\bar{\partial}} \mathcal{E}^{p,q+2} \dots.$$

where the vector spaces in the exact sequence are germs of differential forms. A better formulation is obtained using sheaf theory.

4 Hilbert spaces

On a Riemann surface S introduce a metric, in local coordinates (x, y) by,

$$\lambda(z)(dz) = \lambda(z)\sqrt{dx^2 + dy^2},$$

where $\lambda^2(z) > 0$. In order to have an intrinsic definition, we impose that, changing in the local coordinates $(\tilde{x}, \tilde{y})$, we have

$$\tilde{\lambda}(\tilde{z}) = \lambda(z) \frac{\partial z}{\partial \tilde{z}}.$$

The scalar product between two vectors $v = (\dot{x}_1(0), \dot{y}_1(0))$ and $w = (\dot{x}_2(0), \dot{y}_2(0))$ is

$$\lambda^2(z)(x_1 x_2 + y_1 y_2).$$

The length of a smooth curve $\gamma : [a, b] \rightarrow S$, $\gamma(t) = (x(t), y(t))$ is

$$l(\gamma) = \int_{\gamma} \lambda(z)(dz) = \int_{[a,b]} \sqrt{\dot{x}^2 + \dot{y}^2} dt.$$

We define the distance between two points, z_1 and z_2 to be

$$\inf\{l(\gamma) \mid \gamma : [0, 1] \rightarrow S, \gamma(0) = z_1, \gamma(1) = z_2\}.$$

Definition 4.1 *Define the operator $*$: $\Lambda^k \rightarrow \Lambda^{2-k}$ on functions, 1-forms and 2-forms, in local coordinates (U, z) , as follows*

1. $*f(z) = f(z)\lambda^2(z)dz \wedge d\bar{z}$.
2. $*(f(z)dz + g(z)d\bar{z}) = -if(z)dz + ig(z)d\bar{z}$.
3. $*(f(z)dz \wedge d\bar{z}) = f(z)/\lambda^2(z)$.

Observe that acting on k -forms, $** = (-1)^k$. The geometric interpretation of the $*$ -operator acting on 1-forms is given by the formula $(*df)(v) = df(Jv)$. That is the dual of the J -operator acting on vectors.

Define measurable 2-forms, locally, as given by $f(z)dz \wedge d\bar{z}$ where $f(z)$ is a measurable function. A measurable 1-form is locally given by

$$\alpha(z) = \varphi(z)dz + \psi(z)d\bar{z}$$

with measurable coefficients $\varphi(z)$ and $\psi(z)$. We will identify two forms if they differ in a measure zero set. Observe that measure zero sets are well defined on a manifold because the change of coordinates are given by diffeomorphisms.

Given two measurable k-forms we define the product

Definition 4.2

$$(\alpha_1, \alpha_2) = \int_S \alpha_1 \wedge * \bar{\alpha}_2$$

If α_1 and α_2 are two measurable 1-forms, given in local charts by $\alpha_1(z) = \varphi_1(z)dz + \psi_1(z)d\bar{z}$ and $\alpha_2 = \varphi_2(z)dz + \psi_2(z)d\bar{z}$,

$$\alpha_1 \wedge * \bar{\alpha}_2(z) = i(\varphi_1(z)\bar{\varphi}_2(z) + \psi_1(z)\bar{\psi}_2(z))dz \wedge d\bar{z} = 2(\varphi_1(z)\bar{\varphi}_2(z) + \psi_1(z)\bar{\psi}_2(z))dx \wedge dy.$$

So we see that the product is positive but it might be infinite. Define

$$L_k^2 = \{ \alpha \text{ k-form such that } (\alpha, \alpha) < \infty \}.$$

Lemma 4.3 L_k^2 is a Hilbert space with Hermitian product $(\cdot, \cdot)$.

Proof. Clearly, the product is bilinear and positive definite by the formula above. Also, in the case of 1-forms,

$$(\alpha_1, \alpha_2) = \int_S 2(\varphi_1(z)\bar{\varphi}_2(z) + \psi_1(z)\bar{\psi}_2(z))dx \wedge dy = \overline{(\alpha_2, \alpha_1)}.$$

So $(\cdot, \cdot)$ is Hermitian. To prove completeness we refer to the proof that square integrable functions form a Hilbert space. $\square$

Exercise 4.4 Define the operator $d^* = (- * d *)$ acting on smooth 1-forms.

1. Prove that on smooth forms over a compact surface we have the following formula

$$(d\alpha_1, \alpha_2) = (\alpha_1, (- * d *) \alpha_2)$$

2. Prove that for a function $f : S \rightarrow \mathbb{C}$,

$$d^* df = -\Delta f = -\frac{4}{\lambda^2(z)} \frac{\partial^2 f}{\partial z \partial \bar{z}}$$

Definition 4.5 A harmonic 1-form α is a form satisfying

$$d\alpha = d^* \alpha = 0.$$

Definition 4.6 A holomorphic 1-form α is a form such that, in local coordinates,

$$\alpha = f(z)dz.$$

with $f(z)$ holomorphic.

Exercise 4.7 1. Prove that α is harmonic if and only if, locally, $\alpha = df$ with f harmonic.

2. Prove that α is holomorphic if and only if, locally, $\alpha = df$ with f holomorphic.

Let S be a Riemann surface. Define

$$B = \text{closure of } \{df \mid f \text{ is a smooth function with compact support} \}$$

and

$$B^* = \text{closure of } \{*df \mid f \text{ is a smooth function with compact support} \}.$$

And let

$$B^\perp = \{ \alpha \in L_1^2 \mid (\alpha, df) = 0, f \in C_0^\infty \}$$

and

$$B^{*\perp}$$

be their orthogonals in L_1^2 . By Hilbert space theory, $B^\perp$ and $B^{*\perp}$ are closed.

Proposition 4.8 If α is of class C^1 then

1. $\alpha \in B^\perp$ if and only if $d\alpha = 0$.

2. $\alpha \in B^{*\perp}$ if and only if $d*\alpha = 0$.

A simple consequence of the proposition above is the following

Proposition 4.9

$$L_1^2 = B \oplus B^* \oplus (B^\perp \cap B^{*\perp}).$$

We note $H = B^\perp \cap B^{*\perp}$.

Theorem 4.10 The Hilbert space $B^\perp \cap B^{*\perp}$ is the space of harmonic differentials in L_1^2 .

Proof. If the form is harmonic and in L_1^2 then it is closed and co-closed and therefore in $B^\perp \cap B^{*\perp}$. Reciprocally, suppose that $\alpha \in H$. Choose a neighborhood $U \subset S$ and a function φ with compact support in U . Write, in local coordinates, $\alpha = \alpha_1 dx + \alpha_2 dy$ and $d\varphi = \varphi_1 dx + \varphi_2 dy$. We have

$$(\alpha, d\varphi_1) = 0 = \int_S (\alpha_1 \varphi_{11} + \alpha_2 \varphi_{12}) dx \wedge dy$$

and

$$(\alpha, *d\varphi_2) = 0 = \int_S (-\alpha_1 \varphi_{22} + \alpha_2 \varphi_{21}) dx \wedge dy.$$

Therefore

$$0 = (\alpha, d\varphi_1 - *d\varphi_2) = \int_S (\alpha_1 \varphi_{11} + \alpha_1 \varphi_{22}) dx \wedge dy.$$

By Weyl's lemma we conclude that α_1 is harmonic. Analogously, using $*\alpha$, we conclude that α_2 is harmonic. The result follows.

□

4.1 Smoothing operators

In this section we show Weyl's lemma. Let $\Omega \in \mathbb{R}^2$ be a bounded open set. We will assume known the fact that $C^0(\Omega) \cap L^2(\Omega) \in L^2(\Omega)$ is a dense inclusion. We start now proving that $C^\infty(\Omega) \in L^2(\Omega)$ is a dense inclusion. For this sake extend a function $u \in L^2(\Omega)$ to $L^2(\mathbb{R}^2)$ by defining it to be null on the complement of Ω . introduce the smoothing operator $S_h : L^2(\Omega) \rightarrow C^\infty(\mathbb{R}^2)$ given by

$$u \rightarrow u_h(x) = \frac{1}{2} \int_{\mathbb{R}^2} \rho\left(\frac{x-y}{h}\right) u(y) dy$$

where $\rho \in C^\infty$ is a non-negative function with compact support in D with $\int_D \rho(y) dy = 1$.

Lemma 4.11 $u_h \in C_0^\infty(\mathbb{R}^2)$.

Proof. The integral is well defined because

$$\int_{\mathbb{R}^2} |\rho\left(\frac{x-y}{h}\right) u(y)| dy \leq \int_{\mathbb{R}^2} (\rho\left(\frac{x-y}{h}\right))^2 dy \int_{\mathbb{R}^2} |u(y)|^2 dy$$

One shows that it is smooth by observing that we can derive under the integral. As $u(y) = 0$ outside a compact set, for a fixed h and x sufficiently large the integral is null. $\square$

In particular, the restriction of u_h to Ω is also in $L^2(\Omega)$. Observe that, by a change of variable, one can write

$$u_h(x) = \int_{|z|<1} \rho(z) u(x - hz) dz.$$

We obtain, using Schwarz inequality first,

$$\int_{\Omega} |u_h(x)|^2 dx \leq \int_{\Omega} \left(\int_{|z|<1} \rho(z) dz \int_{|z|<1} \rho(z) |u(x - hz)|^2 dz \right) dx = \int_{\Omega} \int_{|z|<1} \rho(z) |u(x - hz)|^2 dz dx$$

and therefore, using Fubini first and then a change of variables,

$$= \int_{|z|<1} \rho(z) \left(\int_{\Omega} |u(x - hz)|^2 dx \right) dz \leq \int_{\Omega} |u(x)|^2 dx.$$

Proposition 4.12 1. If $u \in C^0(\Omega)$ then for, each compact $K \subset \Omega$,

$$\lim_{h \rightarrow 0} \sup_{x \in K} |u(x) - u_h(x)| = 0$$

2. If $u \in L^2(\Omega)$ then

$$\lim_{h \rightarrow 0} \|u - u_h\|_{L^2(\Omega)} = 0$$

Proof. For the first item, remark that

$$\sup_{x \in K} |u(x) - u_h(x)| \leq \sup_{x \in K} \int_{|z|<1} \rho(z) |u(x) - u(x - hz)| dz$$

so, for $h < \text{dist}(K, \partial\Omega)$

$$\leq \sup_{x \in K} \sup_{|z|<1} |u(x) - u(x - hz)|.$$

But u is uniformly continuous on compact sets and therefore u_h converges uniformly to u on K .

For the second item we write

$$\|u - u_h\|_{L^2(\Omega)} \leq \|u - v\|_{L^2(\Omega)} + \|v - v_h\|_{L^2(\Omega)} + \|v_h - u_h\|_{L^2(\Omega)},$$

where v is a continuous function near u in the L^2 norm. Using the first item in the second term of the right side and $\|v_h - u_h\|_{L^2(\Omega)} \leq \|v - u\|_{L^2(\Omega)}$ as computed in the paragraph above, we obtain the result. $\square$

4.2 Weyl's lemma

Theorem 4.13 *Let $u \in L^2(\Omega)$ such that, for all $\varphi \in C_0^\infty(\Omega)$,*

$$\int_{\Omega} u(x) \Delta \varphi(x) dx = 0.$$

Then u is harmonic.

Proof. Consider the smothings u_h defined in the last section. We have that for small h ,

$$\Delta u_h(x) = \int_{\Omega} \Delta_x \rho_h(x - y) u(y) dy = - \int_{\Omega} \Delta_y \rho_h(x - y) u(y) dy = 0$$

by hypothesis, as $\rho_h(x - y)$ has compact support for sufficiently small h . So $u_h(x)$ is harmonic and by previous results it also is uniformly bounded in $L^1(\Omega)$. We need the following lemma to conclude the proof:

Lemma 4.14 *Let u_n be a sequence of harmonic functions which are uniformly bounded in $L^1(\Omega)$. Then there exists a subsequence which converges uniformly on compact subsets to a harmonic function.*

$\square$

4.3 Poisson equation

Theorem 4.15 *Let $\Omega \subset \mathbb{C}$ be If $\mu \in C_0^\infty(\Omega)$ then the equation*

$$\Delta \varphi = \mu$$

has a C^∞ solution.

Proof. Extend μ to be 0 outside Ω and consider the equation on $\mathbb{C}$. We will show that a smooth solution is given by

$$\varphi(z) = \frac{1}{2\pi} \int \int \log |\zeta| \mu(z + \zeta) \frac{d\zeta \wedge d\bar{\zeta}}{-2i}.$$

The integral is well defined; the only problem being at the origin $\zeta = 0$. We use polar coordinates and write the integrand as $\mu(z + re^{i\theta}) r \log r d\theta dr$ which is clearly integrable

at the origin as $\lim_{r \rightarrow 0} r \log r = 0$. By differentiating the integrand we also obtain that $\varphi \in C^\infty(\Omega)$.

By a change of variable $u = z + \zeta$ we have

$$\varphi(z) = \frac{1}{2\pi} \int \int \log |u - z| \mu(u) \frac{du \wedge d\bar{u}}{-2i}.$$

Therefore (writing in polar coordinates one sees, as above, that the following integral is well defined)

$$\begin{aligned} \frac{\partial \varphi(z)}{\partial z} &= \frac{1}{2\pi} \int \int \frac{\partial \log |u - z|}{\partial z} \mu(u) \frac{du \wedge d\bar{u}}{-2i} = \frac{1}{2\pi} \int \int \frac{-1/2}{u - z} \mu(u) \frac{du \wedge d\bar{u}}{-2i} \\ &= -\frac{1}{4\pi} \int \int \frac{1}{\zeta} \mu(z + \zeta) \frac{d\zeta \wedge d\bar{\zeta}}{-2i}. \end{aligned}$$

We compute now the Laplacian

$$\begin{aligned} \Delta \varphi(z) &= 4 \frac{\partial^2 \varphi}{\partial \bar{z} \partial z}(z) = -\frac{1}{\pi} \int \int \frac{1}{\zeta} \frac{\partial \mu(z + \zeta)}{\partial \bar{z}} \frac{d\zeta \wedge d\bar{\zeta}}{-2i} \\ &= -\frac{1}{\pi} \int \int \frac{1}{u - z} \frac{\partial \mu}{\partial \bar{u}}(u) \frac{du \wedge d\bar{u}}{-2i}. \end{aligned}$$

To compute the last integral, we integrate on a disc of sufficiently large radius and use Cauchy's formula to obtain

$$\mu(z).$$

□

Remark: The function $G(w) = \frac{1}{2\pi} \log |w|$ is called a Green function. Observe that, outside the origin, G is harmonic and, for $\psi \in C_0^\infty(\mathbb{C})$,

$$\int \int G(w) \Delta \psi(w) \frac{du \wedge d\bar{u}}{-2i} =$$

For a smooth function μ we obtained the solution of Poisson equation, $\Delta \varphi = \mu$, as

$$\varphi(z) = \int \int G(z - u) \mu(u) \frac{du \wedge d\bar{u}}{-2i}.$$

Lemma 4.16 *Let S be a Riemann surface. If $\omega \in C^\infty(S) \cap L^2(S)$ then, locally, there exist smooth functions f, g such that*

$$\omega = df + *dg.$$

Proof. Let D_1 be a disc of radius one in a coordinate chart. Let ρ be a function with compact support in $D_{\frac{2}{3}}$ and being identically one in $D_{\frac{1}{3}}$. We consider the form

$$\rho \omega = p dx + q dy.$$

so that $d\rho \omega = \mu dx \wedge dy$. Let g be a solution of the equation

$$\Delta g = \mu.$$

That is, $d * dg = d\rho \omega$ and therefore $d(*dg - \rho \omega) = 0$. By Poincaré's lemma, there exists f such that $*dg - \rho \omega = df$. □

Theorem 4.17 *Let S be a Riemann surface. If $\omega \in C^\infty(S) \cap L^2(S)$ then there exist $f, g \in C^\infty(S)$ such that*

$$\omega = \omega_h + df + *dg.$$

Proof. By the previous lemma we can write, in an open set U ,

$$\omega = df + *dg.$$

By the Hodge decomposition

$$\omega = \omega_h + \alpha + \beta.$$

Therefore, on U , we can write

$$\theta = \omega_h + \alpha - df = -\beta + *dg.$$

We claim that θ is harmonic on U and therefore α and β are smooth. To prove the claim we compute, for $h \in C_0^\infty(U)$

$$(\theta, dh)_U = (-\beta, dh)_U + (*dg, dh)_U = (*dg, dh)_U$$

as $(-\beta, dh)_U = (-\beta, dh)_S = 0$. Now

$$(*dg, dh)_U = - \int \int_U dg \wedge d\bar{h} = - \int \int_U \bar{h} ddg = 0.$$

Analogously, we obtain $(\theta, *dh)_U = 0$. □

Remark: In particular, if ω is smooth and closed then there exists a harmonic form and a smooth function f on S such that

$$\omega = \omega_h + df.$$

5 Harmonic and Holomorphic differentials

On a compact Riemann surface, a harmonic form which is exact vanishes identically. On the other hand, exact harmonic forms can be obtained if we allow singularities. In this section we state existence theorems for harmonic and holomorphic differentials with prescribed singularities.

Let $U \subset S$ be a neighborhood in a Riemann surface defined in local coordinates by $|z| < 1$. With a slight abuse of notation, we write a function on U using the local coordinate. For instance, we say that $1/z^n$ is a function defined on U with a singularity at $z_0 = 0$.

Theorem 5.1 *Let S be a Riemann surface and $z_0 \in S$ any point. Fix $n \geq 1$. There exists a 1-form ω , satisfying*

1. ω is harmonic and exact on $S \setminus \{z_0\}$.
2. $\omega - d(\frac{1}{z^n})$ is harmonic on $|z| < 1/2$.

Proof. Let $\rho \in C_0^\infty(S)$ with support in U and such that it is the identity on $|z| < 1/2$. We define the differential $\psi = d(\rho/z^n) \in S \setminus \{z_0\}$. Observe that $\psi \in L^2(S)$ and is of type $(1, 0)$ on $|z| < 1/2$. Therefore the form $\psi - i * \psi$ is smooth on S . By Theorem 4.17 there exist smooth functions f and g such that

$$\psi - i * \psi = \omega_h + df + *dg.$$

Define now $\omega = \psi - df = \omega_h + *dg + i\psi$.

1. Clearly, ω is smooth and exact on $S \setminus \{z_0\}$. To show that it is harmonic, we compute

$$d\omega = d(\psi - df) = 0$$

and

$$d * \omega = d * (\omega_h + *dg + *i\psi) = -id\psi = 0.$$

2. To show that we have the prescribed singularity, observe that on $|z| < 1/2$ $\omega - d(\frac{1}{z^n}) = \omega - \psi = -df$, therefore smooth and closed. Also, $\omega - d(\frac{1}{z^n}) = \omega_h + *dg - (\psi - i * \psi) = \omega_h + *dg$, therefore smooth and co-closed.

□

Remarks:

1. If S is compact, it follows immediately that ω is unique. In fact if ω' satisfies the same conditions we obtain that $\omega - \omega'$ is harmonic and therefore null.
2. For a non-compact surface one can prove the existence and uniqueness with the further condition that $\omega \in L^2(S \setminus U)$.
3. Taking real and imaginary parts of ω we obtain real harmonic differentials with singularities $\alpha_1 = \operatorname{Re} 1/z^n$ and $\alpha_2 = \operatorname{Im} 1/z^n$ respectively.
4. From harmonic differentials we obtain meromorphic differentials. For instance, a meromorphic differential is $\alpha_1 + i * \alpha_1$. It has $d(\frac{1}{z^n})$ as singularity but only its real part is in general exact.

The differentials obtained so far have null residue. In order to obtain differentials with a prescribed residue, consider as before $U \subset S$ a neighborhood in a Riemann surface defined in local coordinates by $|z| < 1$. Let z_1, z_2 such that $|z_i| < 1/2$. Define the meromorphic differential

$$d(\log \frac{z - z_1}{z - z_2}) = \frac{dz}{z - z_1} - \frac{dz}{z - z_2}.$$

Observe that, in fact, $\log \frac{z - z_2}{z - z_1}$ is well defined for $|z| > 1/2$ (prove that).

The following theorem follows with the same proof as the theorem above.

Theorem 5.2 *Let S be a Riemann surface and $z_i \in U \subset S$ two points with $|z_i| < 1/2$. There exists a 1-form ω , satisfying*

1. ω is harmonic (or holomorphic) on $S \setminus \{z_1, z_2\}$ and exact on $S \setminus \{|z| < 1/2\}$.
2. $\omega - d(\log \frac{z - z_1}{z - z_2})$ is harmonic on $\{|z| < 1/2\}$.

Now we can consider any two points z_1, z_2 on a Riemann surface S . Take a path joining the two points and a finite covering U_α by discs of radius one such that the discs of radius $1/2$ contained in them also cover the path. We can apply the theorem above for a sequence of points starting at z_1 and ending at z_2 such that each adjacent pair is contained in a disc of radius $1/2$. By construction we obtain the following

Corollary 5.3 *Let S be a Riemann surface and $z_i \in S$ two points. Then there exists a 1-form ω satisfying*

1. ω is harmonic (or holomorphic) on $S \setminus \{z_1, z_2\}$.
2. ω has singularities $\frac{dz}{z}$ and $-\frac{dz}{z}$ around z_1 and z_2 respectively.

A further construction gives

Corollary 5.4 *Let S be a Riemann surface and $z_i \in S$ any chosen n points. Suppose $c_i \in \mathbb{C}$ satisfy $c_1 + \cdots + c_n = 0$. Then there exists a 1-form ω satisfying*

1. ω is harmonic (or holomorphic) on $S \setminus \{z_i\}$.
2. ω has singularities $c_i \frac{dz}{z}$ around z_i .

Proof. To prove the corollary, we choose another point z_0 (distinct from the first sequence) and apply the previous corollary for each pair (z_i, z_0) with singularities $c_i \frac{dz}{z}$ and $-c_i \frac{dz}{z}$ around z_i and z_0 respectively. We obtain a form ω_i for each pair and $\omega = \omega_1 + \cdots + \omega_n$ gives the 1-form we need. $\square$

5.1 Meromorphic functions

Meromorphic functions are obtained as quotients of meromorphic differentials. In fact, consider $\omega_1 = g_1(z)dz$ and $\omega_2 = g_2(z)dz$ two meromorphic differentials written in local coordinates. Then $f = g_1(z)/g_2(z)$ is a meromorphic function.

Theorem 5.5 *There exists a non-constant meromorphic function on any Riemann surface.*

Proof. We have to construct two meromorphic functions which are not proportional. To this end, let z_0, z_1, z_2 be three points and construct

1. ω_1 with residue 1 at z_1 and residue -1 at z_0 ,
2. ω_2 with residue 1 at z_2 and residue -1 at z_0 .

The quotient ω_1/ω_2 has certainly a pole at z_1 (maybe of order greater than one) and a zero at z_2 (maybe of higher order). In any case it cannot be constant! $\square$

5.2 The Dirichlet principle

6 Periods and Bilinear Relations

Recall that the period map of closed form α is the map $H^1(M) \rightarrow \mathbb{C}$ given by $\gamma \rightarrow \int_\gamma \alpha$.

Let φ be a closed C^∞ 1-form defined on S and let $a_i, b_i, 1 \leq i \leq g$, be a homology basis.

Definition 6.1 *The a -periods and b -periods of a closed C^∞ 1-form φ are respectively*

$$A_i(\varphi) = \int_{a_i} \varphi \quad \text{and} \quad B_i(\varphi) = \int_{b_i} \varphi$$

In the last section we showed that $H^1(S, \Omega^{1,0})$ has dimension g . Let α_p , $1 \leq p \leq g$, be a basis of $H^1(S, \Omega^{1,0})$.

We define the period matrix of S as the $g \times 2g$ matrix with columns (called vector periods)

$$P_i = (A_1(\alpha_i), \dots, A_g(\alpha_i)), \quad P_{i+g} = (B_1(\alpha_i), \dots, B_g(\alpha_i)).$$

Lemma 6.2 *The vectors $P_j \in \mathbb{C}^g$ are linearly independent over $\mathbb{R}$.*

Proof. Otherwise, there would exist a linearly combination $\alpha = \sum c_i \alpha_i$ with zero periods and therefore $\alpha = 0$. $\square$

This implies that the set of vector periods P_i define a lattice $\Lambda \subset \mathbb{C}^g$.

Definition 6.3 *The Jacobian variety $J(S)$ is the complex torus $\mathbb{C}^g/\Lambda$.*

The Jacobian map

$$j_{z_0} : S \rightarrow J(S)$$

is given by

$$j(z) = \left(\int_{z_0}^z \alpha_1, \dots, \int_{z_0}^z \alpha_g \right)$$

where z_0 is a chosen point in S and the integrals are computed using any path. The map is well defined because different choices of paths will lead to equal vectors modulo Λ .

Proposition 6.4 *Let φ_1 and φ_2 be two closed 1-forms defined on S . Then*

$$\int \int_S \varphi_1 \wedge \varphi_2 = \sum_i (A_i(\varphi_1)B_i(\varphi_2) - B_i(\varphi_1)A_i(\varphi_2))$$

Proof. Let Δ be the polygon whose boundary is a homology basis. Fix a point $z_0 \in \text{int}(\Delta)$. If α is closed and defined on Δ we can define for each $P \in \Delta$,

$$u(P) = \int_{z_0}^P \alpha$$

Suppose that φ is a 1-form defined on a neighborhood of $\partial\Delta$. The proposition follows immediately from the following lemma.

Lemma 6.5 *With the same notation as above*

$$\int_{\partial\Delta} u\varphi = \sum_i (A_i(\alpha)B_i(\varphi) - B_i(\alpha)A_i(\varphi))$$

Proof. Write

$$\int_{\partial\Delta} u\varphi = \int_{a_i} u\varphi + \int_{-a_i} u\varphi + \int_{b_i} u\varphi + \int_{-b_i} u\varphi$$

Observe now that, for corresponding points in z and z' in a_i and $-a_i$,

$$u(z') - u(z) = \int_{z_0}^{z'} \alpha - \int_{z_0}^z \alpha = \int_z^{z'} \alpha = \int_z^{z'} \alpha = \int_{b_i} \alpha$$

and therefore

$$\int_{a_i} u\varphi + \int_{-a_i} u\varphi = \int_{a_i} (u(z) - u(z'))\varphi(z) = - \int_{b_i} \alpha \int_{a_i} \varphi.$$

Analogously, we have

$$\int_{b_i} u\varphi + \int_{-b_i} u\varphi = \int_{a_i} \alpha \int_{b_i} \varphi.$$

□

□

Observe, from the proof, that φ_2 need to be defined only on a neighborhood of the homology basis. Another way to write this is with the help of the matrix

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

and the vector of periods $P(\varphi) = (A_1(\varphi), \dots, B_1(\varphi), \dots)$, so that

$$\int_S \varphi_1 \wedge \varphi_2 = P(\varphi_1)JP^T(\varphi_2)$$

We apply this result to the case of φ_1 holomorphic and φ_2 meromorphic.

Theorem 6.6 *Let φ_1 (holomorphic) and φ_2 (meromorphic) be two 1-forms defined on S . Suppose φ_2 is nonsingular along the homology basis and let $u = \int_{z_0}^z \varphi_1$ for $z_0 \in \Delta$. Then*

$$2\pi i \sum \text{Res}(u\varphi_2) = \sum_i (A_i(\varphi_1)B_i(\varphi_2) - B_i(\varphi_1)A_i(\varphi_2))$$

Proof. The proof follows from the previous proposition and the identity:

$$2\pi i \sum \text{Res}(u\varphi_2) = \int_{\partial\Delta} u\varphi_2 = \int \int_S \varphi_1 \wedge \varphi_2$$

□

Remark. Suppose that at $z = 0$ we have

$$\varphi_1(z) = (a_0 + \dots)dz$$

and

$$\varphi_2(z) = (b_{-m}z^{-m} + \dots + b_0 + \dots)dz.$$

Then

$$u(z) = a_0z + \dots$$

and at $z = 0$

$$\text{Res}_0(u\varphi_2) = \sum_{k=2}^m \frac{a_{k-2}}{k-1} b_{-k}.$$

If both forms φ_i are holomorphic we obtain

Lemma 6.7 *Let φ_1 and φ_2 be two holomorphic 1-forms defined on S . Then*

$$\sqrt{-1} \int \int_S \varphi_1 \wedge \bar{\varphi}_2 = \sqrt{-1} \sum_i (A_i(\varphi_1) B_i(\bar{\varphi}_2) - B_i(\varphi_1) A_i(\bar{\varphi}_2))$$

Observe that, if $\varphi_1 = \varphi_2 = \varphi$, the lemma implies that

$$0 < (\varphi, \varphi) = \sqrt{-1} \int \int_S \varphi \wedge \bar{\varphi} = \sqrt{-1} \sum_i (A_i(\varphi) B_i(\bar{\varphi}) - B_i(\varphi) A_i(\bar{\varphi})).$$

Theorem 6.8 (Riemann's bilinear relations) *Let S be a compact surface of genus g . Then there exists a basis of holomorphic differentials such that the period matrix has the form*

$$(I, Z)$$

where I is the identity matrix of order g and Z a symmetric matrix satisfying $\text{Im} Z > 0$.

Proof. A change of basis of holomorphic forms is given by a matrix $g = (g_j^i)$ so that $\varphi_j = \sum_i g_j^i \varphi_i$. In that case, the new period matrix π' is $\pi' = g\pi$. We need to prove that π_{ij} , $1 \leq i, j \leq g$ is invertible. But if this not the case there would exists a linearly combination of holomorphic forms, say φ with zero a -periods. From the formula before the theorem this is impossible.

Now, in that basis $A_i(\varphi_l) = \delta_{il}$ and by Lemma 6.6 we obtain

$$0 = \sum_i (A_i(\varphi_l) B_i(\varphi_m) - B_i(\varphi_l) A_i(\varphi_m))$$

which implies that $B_l(\varphi_m) = B_m(\varphi_l)$ so that Z is symmetric.

Lastly, using Lemma 6.7 we get

$$0 < (\varphi_i, \varphi_j) = \sqrt{-1} \int \int_S \varphi_i \wedge \bar{\varphi}_j = \sqrt{-1} (B_i(\bar{\varphi}_j) - B_j(\varphi_i)) = 2\text{Im } B_j(\varphi_i).$$

□

Exercises. Let S be a compact Riemann surface.

1. (normalized abelian differentials of the third kind) Prove that there exists a unique meromorphic 1-form, ω_{z_1, z_2} with only two simple poles with residues $+1$ and -1 at two points z_1, z_2 such that its a -periods are null.
2. (normalized abelian differentials of the second kind) Prove that there exists a unique meromorphic 1-form, ω_{z_0} with only one pole at z_0 such that at a local coordinate neighborhood (U, z) around z_0 , $\omega - \frac{dz}{z^n}$ is holomorphic and such that its a -periods are null.
3. Any meromorphic 1-form is a combination of holomorphic 1-forms, abelian differentials of the third kind and abelian differentials of the second kind.
4. Prove the reciprocity relations (φ_k is a normalized basis of holomorphic forms)

$$\int_{b_k} \omega_{z_1, z_2} = 2\pi i \int_{z_2}^{z_1} \varphi_k$$

and ($\varphi_k = f_k(z)dz$ on U)

$$\int_{b_k} \omega_{z_0} = 2\pi i \frac{f_k^{n-1}(z_0)}{n!}.$$

7 Riemann-Roch Theorem

We will suppose the Riemann surface S compact in this section. Recall that a principal divisor is a divisor (f) associated to a meromorphic function f . A canonical divisor is a divisor $K = (\omega)$ associated to a meromorphic 1-form ω . As the quotient of two meromorphic forms is a meromorphic function we have

Lemma 7.1 *Any two canonical divisors are linearly equivalent.*

Proposition 7.2 $\deg K = 2g - 2$.

Definition 7.3 *A meromorphic function (or abelian differential) g is a multiple of a divisor D if it is zero or*

$$D + (g) \geq 0.$$

We write, for a divisor D on S ,

$$|D| = \{ D' \in \text{Div}(S) \mid \text{effective and linearly equivalent to } D \}.$$

This is not a vector space but is related to one;

Definition 7.4 *Define the space*

$$\mathcal{L}(D) = \{ \text{meromorphic functions which are multiples of } D \}$$

and its dimension $h^0(D) = \dim_{\mathbb{C}} \mathcal{L}(D)$.

It is easy to see that $\mathcal{L}(D)$ is a vector space. In fact, observe that if $\text{ord}_z f \geq p$ and $\text{ord}_z g \geq p$ then $\text{ord}_z(f + g) \geq p$.

Observe that if $D' = D + (g)$, then (g) is determined up to a multiplicative constant. From the definitions, if $h^0(D) = 0$ if and only if $|D|$ is empty. On the other hand if $h^0(D) \geq 1$, the space $|D|$ is the projective space defined by $L(D) \setminus \{0\}$, so that

$$h^0(D) = \dim_{\mathbb{C}} |D| + 1.$$

Lemma 7.5 *If $\deg D < 0$ then $h^0(D) = 0$.*

Proof. If $f \in \mathcal{L}(D)$ is non-zero we have $f \geq -D$ and so $\deg(f) \geq -\deg(D) > 0$. But the degree of any principal is zero. $\square$

Definition 7.6 *Define the vector space*

$$\Omega(D) = \{ \text{abelian differentials which are multiples of } -D \}$$

and its dimension $i(D) = \dim_{\mathbb{C}} \Omega(D)$ (the index of speciality).

Theorem 7.7 *If S is a Riemann surface of genus g and D any divisor, then*

$$h^0(D) = \deg(D) - g + 1 + i(D) = \deg(D) - g + 1 + h^0(K - D).$$

Remarks: $h^0(K - D) = i(D)$ is the number of linearly independent meromorphic 1-forms ω such that $(\omega) \geq D$. Simply recall that all canonical divisors are linearly equivalent so if K is a canonical divisor $(\omega) = (g) + K$ for a meromorphic function g . We have then $(g) \geq -K + D$ equivalent to $(\omega) \geq D$.

Proof. Suppose first that D is an effective divisor

$$D = \sum l_i z_i \quad 1 \leq i \leq n ,$$

with distinct z_i and $l_i > 0$. Observe that the map $f \rightarrow df$ gives an homomorphism

$$L(D) \rightarrow M_0(D)$$

where $M_0(D)$ is the space of meromorphic forms with no residues and periods and whose poles have order at most $l_i + 1$ at z_i . Its kernel is the space of constant functions. Therefore $l(D) = \dim M_0(D) + 1$.

Any principal part at z_i with poles of order between 2 and $l_i + 1$ can be obtained using the l_i 1-forms $\omega_{z_i}^n$, $n = 2, \dots, l_i + 1$ defined before ($\omega_{z_i}^n$ is meromorphic with only one pole of order n at z_i). Consider the map $l : \mathbb{C}^{l_1 + \dots + l_n} \rightarrow \mathbb{C}^g$ given by the b-periods of the normalized differentials;

$$\eta \rightarrow \left(\int_{b_1} \eta, \dots, \int_{b_g} \eta \right).$$

Clearly, $M_0(D) = \text{Ker } l$. Consider a normalized basis (zero a -periods and $\int_{b_k} \omega_l = \delta_{kl}$) of holomorphic differentials, written in local coordinates, as $\omega_k(z) = f_k(z)dz$. By the reciprocity relations we have

$$\int_{b_k} \omega_{z_i}^n = 2\pi\sqrt{-1} \frac{f_k^{n-1}(z_i)}{n!}.$$

So $\sum c_{n,i} \omega_{z_i}^n = 0$ if and only if $\sum c_{n,i} \frac{f_k^{n-1}(z_i)}{n!} = 0$. We conclude that $\text{Ker } l$ is the space of meromorphic forms ω such that $\text{ord}_{z_i}(\omega) \geq l_i$. That is,

$$\text{Ker } l = \Omega(-D).$$

This concludes the proof for effective divisors or, more generally, divisors which are equivalent to an effective divisor (in that case $h^0(D) = \dim I(D) - 1 = 0$). If $K - D$ is equivalent to an effective divisor, we apply the formula

$$h^0(K - D) = \deg(K - D) - g + 1 + h^0(K - (K - D)) = -\deg D + 2g - 2 - g + 1 + h^0(D),$$

and the formula is also verified.

If the divisor D and $K - D$ are not equivalent to an effective divisor then $h^0(K - D) = h^0(D) = 0$. We should prove in that case that $\deg(D) = g - 1$. We write $D = D_1 - D_2$, with D_i effective. Applying Riemann-Roch we obtain

$$h^0(D_1) = \deg(D_1) - g + 1 + h^0(K - D_1) \geq \deg(D_1) - g + 1 = \deg(D) + \deg(D_2) - g + 1.$$

If $\deg(D) > g - 1$ we obtain $h^0(D_1) > \deg(D_2)$. Therefore, there exists a meromorphic function $g \in L(D_1)$ vanishing at all points of D_2 . But then $(g) + D = (g) + D_1 - D_2 \geq D_1 \geq 0$, a contradiction. Analogously, assuming $\deg(D) > g$ we obtain a contradiction. $\square$

7.1 Applications

In this section we prove that any compact Riemann surface is embedded in a projective space. We start with the following proposition.

Proposition 7.8 *If S is a compact surface which is not $\mathbb{CP}^1$ then at each point $z \in S$ there exists a holomorphic 1-form ω with $\omega(p) \neq 0$.*

Proof. If that is not the case, as all meromorphic one forms are given by $g\omega_0$ (where g is meromorphic and ω_0 a fixed holomorphic 1-form) and so holomorphic one forms are $L(K) = \{ g \mid (g) + (\omega_0) \geq 0 \}$, we obtain that $L(K - z) = L(K)$. By Riemann-Roch, $h^0(z) = 1 - g + 1 - g = 2$. This means that there exists a meromorphic function with only one pole at z . This is impossible if the Riemann surface is not $\mathbb{CP}^1$. $\square$

From the proposition it is easy to see that the following proposition makes sense.

Definition 7.9 *Let ω_i be a basis of holomorphic differentials of a surface of genus g . Write $\omega_i(z) = f_i(z)dz$ in local coordinates. The canonical map is the map*

$$i_K : S \rightarrow \mathbb{CP}^{g-1}$$

given by $i_K(z) = [f_1(z), \dots, f_g(z)]$.

Exercise Prove that $h^0(lK) = 0$ for surfaces of genus 0, $h^0(lK) = 1$ for surfaces of genus 1 and $h^0(lK) = (2l - 1)(g - 1)$ for surfaces of genus $g \geq 2$ and $l \geq 2$.

We define maps into higher dimensional projective space by considering the divisors lK and a basis of l -canonical holomorphic forms of $h^0(lK)$:

Definition 7.10 *Let ω_i be a basis of holomorphic l -canonical forms of $h^0(lK)$ of a surface of genus g . Write $\omega_i(z) = f_i(z)dz^l$ in local coordinates. The l -canonical map is the map*

$$i_{lK} : S \rightarrow \mathbb{CP}^{h^0(lK)-1}$$

given by $i_{lK}(z) = [f_1(z), \dots, f_{h^0(lK)}(z)]$.

Theorem 7.11 *For any surface S of genus $g \geq 3$, the map $i_{3K} : S \rightarrow \mathbb{CP}^{h^0(3K)-1} = \mathbb{CP}^{5g-6}$ is an imbedding.*

Proof. In order to prove the theorem we show that $h^0(lK - z_1 - z_2) < h^0(lK - z_1)$ for any $z_1, z_2 \in S$. This ensures the existence of a holomorphic l -form vanishing at z_1 but not in z_2 (in the case $z_1 = z_2$ it implies the existence of a l -form with a simple zero at z_1).

Now we apply Riemann-Roch and conclude. $\square$

In fact one can prove that we can always imbed in $\mathbb{CP}^3$.

8 Abel's Theorem

Recall the Jacobian variety $J(S)$, $\mathbb{C}^g/\Lambda$ of a surfac of genus g and the Jacobian map

$$j_{z_0} : S \rightarrow J(S)$$

is given by

$$j_{z_0}(z) = \left(\int_{z_0}^z \alpha_1, \dots, \int_{z_0}^z \alpha_g \right)$$

where z_0 is a chosen point in S and the integrals are computed using any path. We can extend this definition to a homomorphism between $\mathbb{Z}$ -modules:

$$j_{z_0} : \text{Div}(S) \rightarrow J(S).$$

Let $\text{Div}_0(S)$ be the degree zero divisors. The restriction map

$$j : \text{Div}_0(S) \rightarrow J(S)$$

does not depend on the base point z_0 . Indeed, let $D = \sum_1^n (z_i - p_i)$ be a zero degree divisor. We compute

$$j(D) = \left(\sum \left(\int_{z_0}^{z_i} \alpha_1 - \int_{z_0}^{p_i} \alpha_1 \right), \dots, \sum \left(\int_{z_0}^{z_i} \alpha_g - \int_{z_0}^{p_i} \alpha_g \right) \right)$$

so that

$$j(D) = \left(\sum \int_{z_i}^{p_i} \alpha_1, \dots, \sum \int_{z_i}^{p_i} \alpha_g \right).$$

Theorem 8.1 *Let S be a compact Riemann surface and $j : \text{Div}_0(S) \rightarrow J(S)$ be its Jacobian map. A divisor $D \in \text{Div}_0(S)$ is principal if and only if $j(D) = 0$.*

Proof. First we consider the case $g = 0$. As there are no non-trivial holomorphic forms we consider $J(S) = \{0\}$ and the theorem says that any divisor $D = \sum_1^n (z_i - p_i)$ is a principal divisor.

We start with a lemma which relates the existence of meromorphic functions with prescribed zeros and poles to that of meromorphic 1-forms of the third kind with prescribed residues:

Lemma 8.2 *Let $D = \sum_1^n (z_i - p_i)$ be a divisor. There exists g such that $(f) = D$ if and only if there exist c_n , $1 \leq n \leq g$ such that the form $\varphi = \sum \omega_{z_i p_i} + \sum c_n \omega_n$ have periods in $2\pi i\mathbb{Z}$.*

Proof. From f we obtain $\varphi = \frac{df}{f}$. By the residue theorem the periods of φ are in $2\pi i\mathbb{Z}$. Conversely, given φ we define $f(z) = e^{\int_{z_0}^z \varphi}$, where z_0 is a fixed point. $\square$

We consider therefore the form $\varphi = \sum \omega_{z_i p_i} + \sum c_n \omega_n$. The a-periods are (as $\omega_{z_i p_i}$ are normalized) $A_n = c_n$. The b-periods are, using the reciprocity relations:

$$\int_{b_k} \varphi = \sum 2\pi i \int_{z_i}^{p_i} \omega_k + \sum c_n \int_{b_k} \omega_n.$$

The conditions in the lemma are equivalent to the existence of integers m_k and n_k such that

$$c_k = 2\pi i m_k \quad \text{and} \quad \sum 2\pi i \int_{z_i}^{p_i} \omega_k + \sum c_n \int_{b_k} \omega_n = 2\pi i n_k.$$

That is

$$\sum \int_{z_i}^{p_i} \omega_k + \sum m_n \int_{b_k} \omega_n = n_k,$$

or

$$\sum \int_{z_i}^{p_i} \omega_k = - \sum m_n \int_{b_k} \omega_n + n_k.$$

But those conditions are precisely the conditions of the statement of the theorem. $\square$

In fact we also have

Theorem 8.3 (Jacobi) *The map $j : \text{Div} \rightarrow J(M)$ is surjective.*

Both theorems combined give a description of the space of all divisors:

Theorem 8.4 *The induced map $j : \text{Pic}^0 \rightarrow J(S)$ is an isomorphism of groups.*